# BREAST CARE-MA DIAGNOSTIC DECISION SUPPORT REPORT

Department of Information Technology | Nehru Institute of Engineering and Technology
Explainable Multi-Agent AI System for Ultrasound Modality

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| **Case ID:** | BUSI-NORMAL-234B | **Date & Time:** | 2026-10-31 00:46:13 |
| **Patient Age:** | 48 | **Family History:** | No |
| **Breast Density:** | B - Scattered fibroglandular | **Symptoms:** | Routine screening |

## 1. Quantitative AI Findings & Model Outputs

| Parameter | Result | Clinical Meaning |
|---|---|---|
| AI Classification | **MALIGNANT** | Calibrated Malignant Probability: 97.2% |
| Tumor Detection | Lesion Detected | Relative Area: 44.44% |
| Morphology (Agent 3) | Irregular / Spiculated | Aspect Ratio: 1.42, Solidity: 0.60 |
| Acoustic Shadowing | Present | Acoustic attenuation distal to mass |
| ACR BI-RADS Category | **BI-RADS 5** | **Highly Suggestive of Malignancy (>=85%)** |

## 2. Retrieved Evidence & Clinical Guidelines (Agent 4)

• **High Suspicion of Malignancy (>50% to <95%)** (ACR BI-RADS Atlas 5th Edition (Ultrasound)): Irregular hypoechoic mass with angular margins and posterior acoustic shadowing, highly concerning for invasive ductal carcinoma (IDC).

• **Classic Malignant Presentation (>=95% Probability)** (ACR BI-RADS Atlas 5th Edition / Society of Breast Imaging): Classic presentation: Spiculated margins, taller-than-wide (non-parallel) orientation, hypoechoic with acoustic shadowing, microcalcifications. Malignancy risk exceeds 95%.

• **Moderate Suspicion of Malignancy (>10% to <=50%)** (ACR BI-RADS Atlas 5th Edition / NCCN Guidelines): Solid lesion with indistinct margins or non-parallel orientation without definite spiculation. Malignancy risk spans 10% to 50%.

## 3. Recommended Clinical Follow-up Actions

• Urgent ultrasound-guided core needle biopsy recommended.

• Immediate multidisciplinary breast oncology and surgical consultation.

• Consider contrast-enhanced breast MRI for staging if invasive disease is confirmed.

**REGULATORY & SAFETY DISCLAIMER:** NOTICE: Breast CARE-MA is an AI decision-support research prototype developed for academic evaluation. It does not provide definitive medical diagnosis and must never supersede clinical judgment by a licensed radiologist or oncologist.